

Baidu Releases ERNIE-4.5 with Thinking: Install and Test Locally



Fahd Mirza
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104 views Sep 12, 2025 #ERNIE

This video installs and test ERNIE-4.5-21B-A3B-Thinking model.

The image shows the Hugging Face model page for **baidu/ERNIE-4.5-21B-A3B-Thinking**. The page features a dark theme with a navigation bar at the top containing links for Models, Datasets, Spaces, Docs, Pricing, Log In, and Sign Up. The model card includes a search bar, a list of tags (Text Generation, Transformers, Safetensors, English, Chinese, ernie4_5_moe, ERNIE4.5, conversational), and a license (apache-2.0). It also displays a 'Model card' section with links to Chat, ERNIE Bot, Hugging Face, Baidu, GitHub, ERNIE, Blog, ERNIE4.5, Discord, ERNIE, X, PaddlePaddle, and License (Apache2.0). A 'Downloads last month' graph shows 33,522 downloads. The 'Safetensors' section indicates a model size of 21.8B params and a tensor type of F32 · BF16. The page also includes a 'Model Highlights' section.

This is designed for complex reasoning tasks, supports 128K context length.

The image shows a YouTube video player interface. The search bar at the top contains the text 'ernie'. The video list shows three videos: 'Baidu Releases ERNIE 4.5 : MoE-based A47B and A3B : Install and Test Locally' (2.2K views, 2 months ago), 'Baidu's ERNIE X1.1 Just Arrived - Countless Times Smarter' (1.3K views, Streamed 2 days ago), and 'Baidu ERNIE-4.5-VI -28B-A3B- Vision Model for Special Tasks - Install'.

The image shows a YouTube video player interface. The search bar at the top contains the text 'paddlepaddle'. The video list shows two videos: 'Install PaddlePaddle Locally - Comparison with PyTorch - Latest and Updated' (656 views, 1 month ago) and 'Install FastDeploy Locally to Serve AI Models with High Performance' (1.1K views, 2 months ago).

```
#
# $ conda activate ai
#
# To deactivate an active environment, use
#
# $ conda deactivate
(ai) Ubuntu@0151-dsm-prmx30177:~$ clear
```

```
Ubuntu@0151-dsm-prmx30177:~$ python -m pip install paddlepaddle-gpu==3.2.0 -i https://www.paddlepaddle.org.cn/packages/stable/cu126/
Looking in indexes: https://www.paddlepaddle.org.cn/packages/stable/cu126/
Collecting paddlepaddle-gpu==3.2.0
  Downloading https://paddle-whl.bj.bcebos.com/stable/cu126/paddlepaddle-gpu/paddlepaddle_gpu-3.2.0-cp311-cp311-linux_x86_64.whl (1890.3 MB)
0.0/1.9 GB ? eta -:--:--
```

```
Ubuntu@0151-dsm-prmx30177:~$ python -m pip install fastdeploy-gpu -i https://www.paddlepaddle.org.cn/packages/stable/fastdeploy-gpu-80_90/ --extra-index-url https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
Looking in indexes: https://www.paddlepaddle.org.cn/packages/stable/fastdeploy-gpu-80_90/, https://mirrors.tuna.tsinghua.edu.cn/pypi/web/simple
Collecting fastdeploy-gpu
  Downloading https://paddle-whl.bj.bcebos.com/stable/fastdeploy-gpu-80_90/fastdeploy-gpu/fastdeploy_gpu-2.2.0-py3-none-any.whl (585.0 MB)
0.0/585.0 MB ? eta -:--:--
```

```
Ubuntu@0151-dsm-prmx30177:~$ python -c "import paddle; paddle.utils.run_check(); from fastdeploy.model_executor.ops.gpu import beam_search_softmax; print('FastDeploy ready')"
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/utils/cpp_extension/extension_utils.py:718: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
warnings.warn(warning_message)
Running verify PaddlePaddle program ...
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/pir/math_op_patch.py:219: UserWarning: Value do not have 'place' interface for pir graph mode, try not to use it. None will be returned.
warnings.warn(
I0912 01:02:40.740180 25851 pir_interpreter.cc:1524] New Executor is Running ...
W0912 01:02:40.741744 25851 gpu_resources.cc:114] Please NOTE: device: 0, GPU Compute Capability: 9.0, Driver API Version: 12.4, Runtime API Version: 12.6
I0912 01:02:40.742596 25851 pir_interpreter.cc:1547] pir interpreter is running by multi-thread mode ...
PaddlePaddle works well on 1 GPU.
PaddlePaddle is installed successfully! Let's start deep learning with PaddlePaddle now.
FastDeploy ready
Ubuntu@0151-dsm-prmx30177:~$
```

We are now ready to download and start doing inference with the LLM

```
Ubuntu@0151-dsm-prmx30177:~$ python -m fastdeploy.entrypoints.openai.api_server \
--model baidu/ERNIE-4.5-21B-A3B-Thinking \
--port 8180 --metrics-port 8181 --engine-worker-queue-port 8182 \
--load_choices "default_v1" --tensor-parallel-size 1 \
--max-model-len 131072 --reasoning-parser ernie_x1 \
--tool-call-parser ernie_x1 --max-num-seqs 32
```

We will use **FastDeploy** engine to create an OpenAI API compatible endpoint

```
Ubuntu@0151-dsm-prmx30177:~$ python -m fastdeploy.entrypoints.openai.api_server \
--model baidu/ERNIE-4.5-21B-A3B-Thinking \
--port 8180 --metrics-port 8181 --engine-worker-queue-port 8182 \
--load_choices "default_v1" --tensor-parallel-size 1 \
--max-model-len 131072 --reasoning-parser ernie_x1 \
--tool-call-parser ernie_x1 --max-num-seqs 32
/home/Ubuntu/miniconda3/envs/ai/lib/python3.11/site-packages/paddle/utils/cpp_extension/extension_utils.py:718: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
warnings.warn(warning_message)
Downloading [README.md]: 100%| 7.28k/7.28k [00:01<00:00, 3.84kB/s]
Downloading [LICENSE]: 100%| 11.1k/11.1k [00:01<00:00, 5.89kB/s]
Downloading [config.json]: 100%| 1.11k/1.11k [00:02<00:00, 552B/s]
Processing 16 items: 19%| 3.00/16.0 [00:05<00:24, 1.88s/it]
Downloading [model-00001-of-00009.safetensors]: 0%| 0.00/4.65G [00:00<?, ?B/s]
```

```
EXPLORER
...
app.py
root@16226313822: /workspace

014221 seconds.
INFO 2025-09-12 01:54:49,751 61 api_server.py[line:552] Launching metrics service at http://0.0.0.0:8181/metrics
INFO 2025-09-12 01:54:49,751 61 api_server.py[line:553] Launching chat completion service at http://0.0.0.0:8180/v1/chat/completions
INFO 2025-09-12 01:54:49,751 61 api_server.py[line:554] Launching completion service at http://0.0.0.0:8180/v1/completions
Processing 16 items: 100% | 16.0/16.0 [00:16<00:00, 1.02s/it]
[2025-09-12 01:55:09] [ INFO ] - Started server process [61]
[2025-09-12 01:55:09] [ INFO ] - Waiting for application startup.
[2025-09-12 01:55:09,531] [ INFO ] - Using download source: huggingface
[2025-09-12 01:55:09,532] [ INFO ] - Loading configuration file /root/PaddlePaddle/ERNIE-4.5-21B-A3B-Thinking/config.json
[2025-09-12 01:55:09,532] [ WARNING ] - You are using a model of type ernie4_5_moe to instantiate a model of type . This is not supported for all configurations of models and can yield errors.
[2025-09-12 01:55:09,543] [ INFO ] - Using download source: huggingface
[2025-09-12 01:55:09,543] [ INFO ] - Loading configuration file /root/PaddlePaddle/ERNIE-4.5-21B-A3B-Thinking/config.json
[2025-09-12 01:55:09,543] [ WARNING ] - You are using a model of type ernie4_5_moe to instantiate a model of type . This is not supported for all configurations of models and can yield errors.
[2025-09-12 01:55:09,554] [ INFO ] - Using download source: huggingface
[2025-09-12 01:55:09] [ INFO ] - Application startup complete.
[2025-09-12 01:55:09] [ INFO ] - Uvicorn running on http://0.0.0.0:8180 (Press CTRL+C to quit)
```

```
Ubuntu@0151-dsm-prxm30177: ~/mycode
(ai) Ubuntu@0151-dsm-prxm30177:~/mycode$ nvidia-smi
Fri Sep 12 01:55:36 2025

+-----+
| NVIDIA-SMI 550.54.14                  | Driver Version: 550.54.14      | CUDA Version: 12.4 |
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC | | | |
| Fan  Temp  Perf    Pwr:Usage/Cap       | Bus-Id        Memory-Usage | GPU-Util  Compute M. |
|====|=====|=====|=====|=====|=====|
| 0   NVIDIA H100 PCIe                    On           | 00000000:01:00:0 Off |                    0 |
| N/A   26C    P0              72W / 350W | 00000000:01:00:0 70871MiB / 81559MiB | 0%      Default |
|                                     | I               Disabled |
+-----+-----+

Processes:
+-----+
| GPU   GI   CI        PID   Type   Process name                      | GPU Memory |
|  ID   ID   ID              |              Usage                   |
+-----+-----+
| 0   N/A  N/A       1293    G   /usr/lib/xorg/Xorg                  | 108MiB     |
| 0   N/A  N/A       2354    G   /usr/bin/gnome-shell                | 9MiB       |
| 0   N/A  N/A      28618    C   /usr/bin/python                     | 43226MiB   |
| 0   N/A  N/A      28685    C   /usr/bin/python                     | 27496MiB   |
+-----+-----+

(ai) Ubuntu@0151-dsm-prxm30177:~/mycode$
```

This is consuming close to 71GB of VRAM

```
EXPLORER
...
app.py
MYCODE
app.py

1 import requests
2 import json
3 from rich.console import Console
4 from rich.panel import Panel
5 from rich.markdown import Markdown
6 from rich.text import Text
7
8 console = Console()
9 console.print("[bold blue]Sending request to ERNIE-4.5-21B-A3B-Thinking...[/bold blue]")
10
11 prompt=""
12 Write 'Fahd Mirza' in Chinese
13 ""
14
15 response = requests.post("http://localhost:8180/v1/chat/completions",
16 headers={"Content-Type": "application/json"},
17 json={
18     "model": "baidu/ERNIE-4.5-21B-A3B-Thinking",
19     "messages": [{"role": "user", "content": prompt}],
20     "max_tokens": 1000,
21     "temperature": 0.7
22 })
23
24
25 result = response.json()["choices"][0]["message"]["content"]
26 console.print(Panel(Markdown(result), title="[bold green]ERNIE Response[/bold green]", border_style=
```

We will use this code to talk to the ERNIE LLM on its API endpoint.


```
Ubuntu@0151-dsm-prmx30177: ~/mycode
(ai) Ubuntu@0151-dsm-prmx30177:~/mycode$ python app.py
Sending request to ERNIE-4.5-21B-A3B-Thinking...
ERNie Response
Fahd Mirza 的中文翻译通常为 法赫德·米尔扎 (按照英文发音直接音译得出)。
(ai) Ubuntu@0151-dsm-prmx30177:~/mycode$
```

This is a bilingual model in English and Chinese as seen above

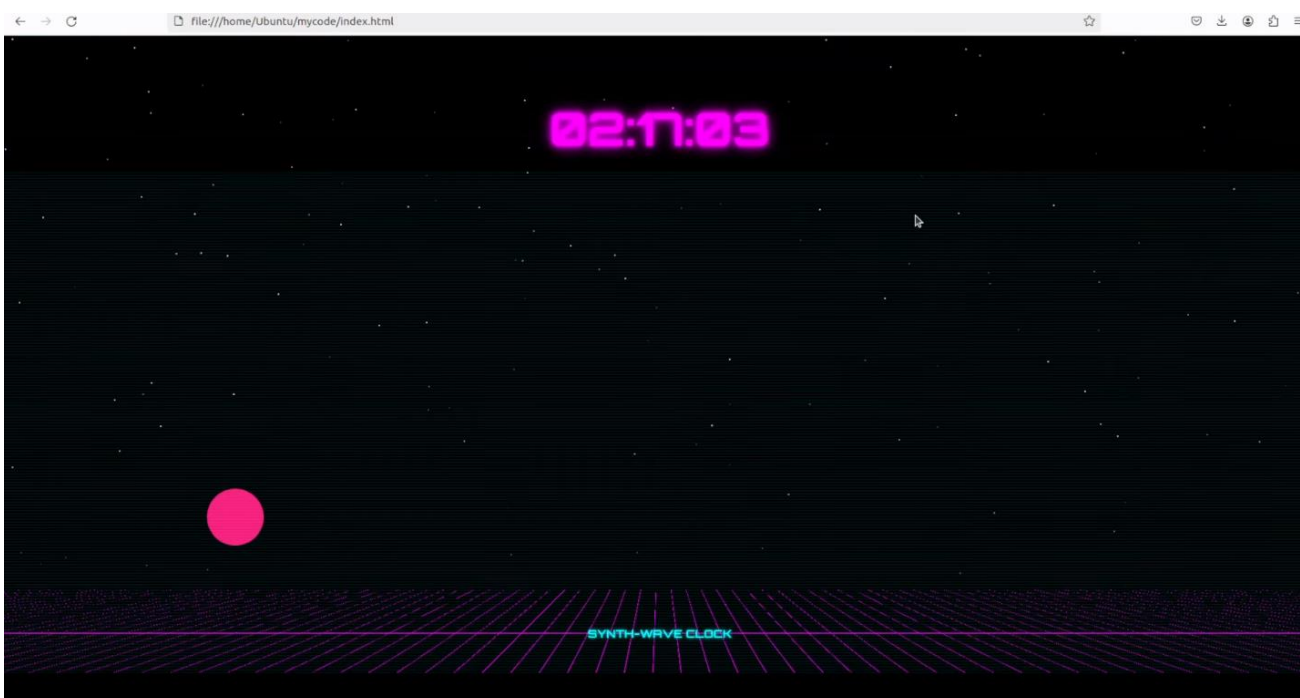
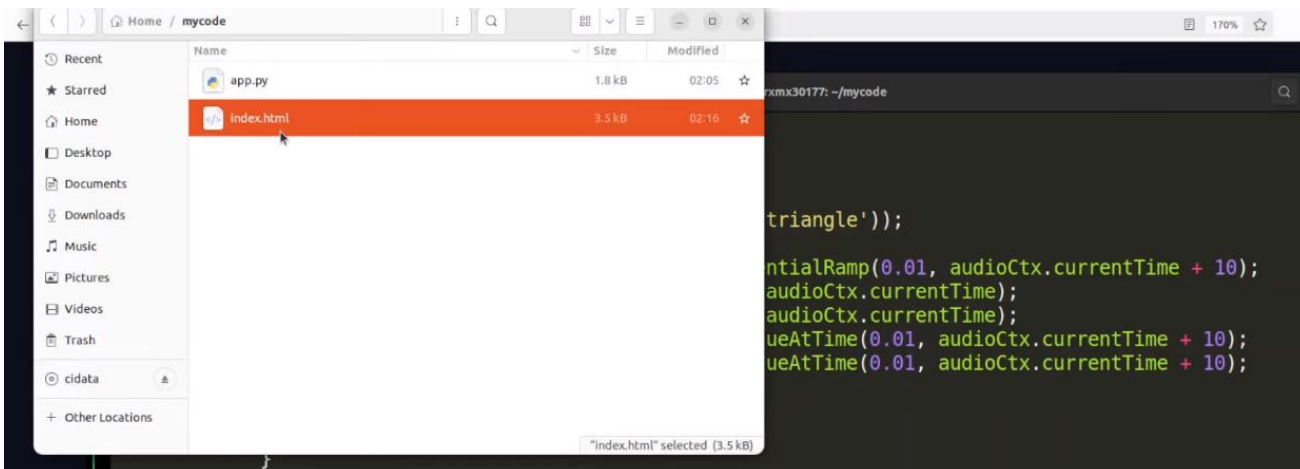
```
EXPLORER
MYCODE
app.py
app.py
1 import requests
2 import json
3 from rich.console import Console
4 from rich.panel import Panel
5 from rich.markdown import Markdown
6 from rich.text import Text
7
8 console = Console()
9 console.print("[bold blue]Sending request to ERNIE-4.5-21B-A3B-Thinking...[/bold blue]")
10
11 prompt="""
12 Create one self-contained HTML file (no external requests) that displays a real-time retro synth-wa
13
14 - Full-screen black background with a neon purple grid horizon at the bottom
15 - A glowing circular sun that moves from bottom-left to bottom-right and back once every 60 seconds
16 - Sun colour animates from hot-pink (bottom-left) to orange-red (bottom-right)
17 - Large neon digital hours:minutes:seconds centered near the top, using a glowing 80s-style font vi
18 - Gentle star-field twinkle behind the grid (tiny white dots, random opacity)
19 - Add a quiet 80s pad drone (Web-Audio: two oscillators through a low-pass filter) that starts auto
20 - Embed the small Web-Audio code inline; no external assets
21 - Return only the raw HTML, no markdown code fences
22 """
23
24 response = requests.post("http://localhost:8180/v1/chat/completions",
25 headers={"Content-Type": "application/json"},
26 json={
27     "model": "baidu/ERNIE-4.5-21B-A3B-Thinking",
28     "messages": [{"role": "user", "content": prompt}],
29     "max_tokens": 1000,
30     "temperature": 0.7
31 })
```

We can now do some inference on the API as above

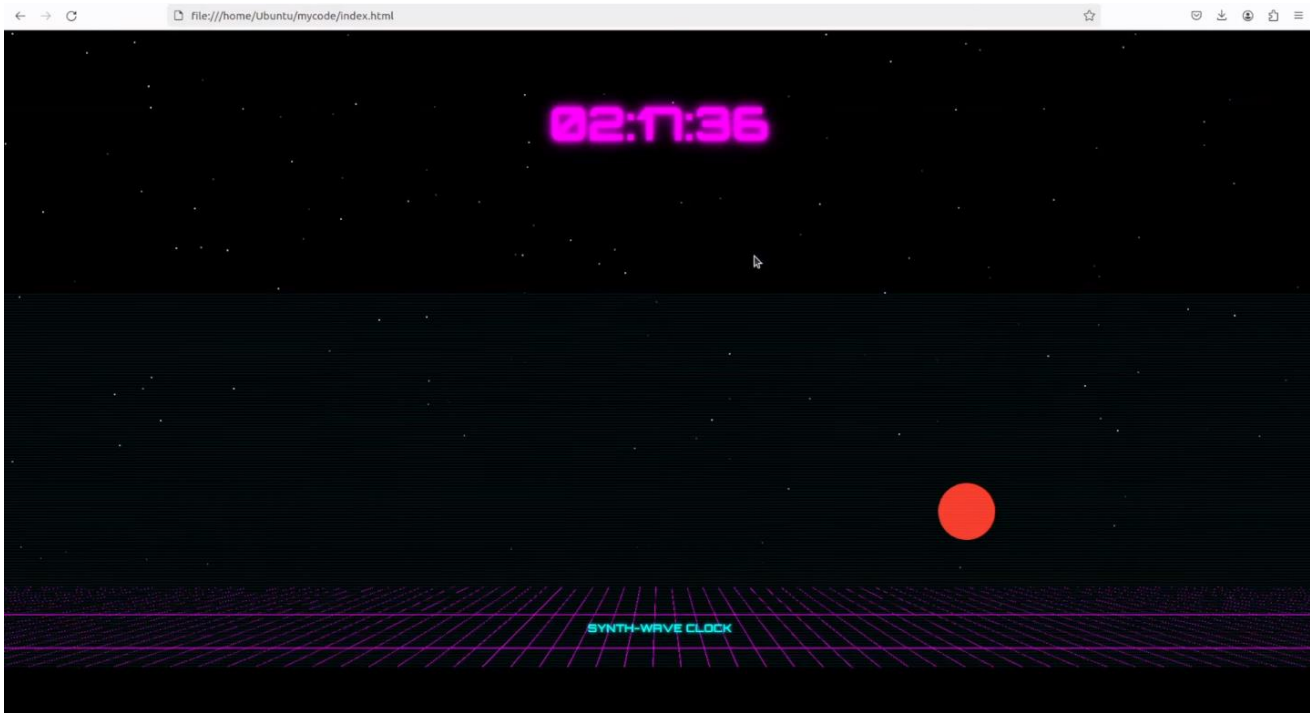
```
EXPLORER
MYCODE
app.py
app.py
1
2
3 import Console
4 import Panel
5 import Markdown
6 import Text
7
8 ()
9 [bold blue]Sending request to ERNIE-4.5-21B-A3B-Thinking...[/bold blue]
10
11
12 ontained HTML file (no external requests) that displays a real-time retro synth-wave sunset clock:|
13
14 :k background with a neon purple grid horizon at the bottom
15 lar sun that moves from bottom-left to bottom-right and back once every 60 seconds, synced to the re
16 ates from hot-pink (bottom-left) to orange-red (bottom-right)
17 tal hours:minutes:seconds centered near the top, using a glowing 80s-style font via CSS text-shadow
18 ld twinkle behind the grid (tiny white dots, random opacity)
19 pad drone (Web-Audio: two oscillators through a low-pass filter) that starts automatically on first
20 Web-Audio code inline; no external assets
21 raw HTML, no markdown code fences
22
23
24 ts.post("http://localhost:8180/v1/chat/completions",
25 tent-Type": "application/json"),
26
27 "baidu/ERNIE-4.5-21B-A3B-Thinking",
28 ": [{"role": "user", "content": prompt}],
29 s": 1000,
30 res": 0.7
```

```
EXPLORER
MYCODE
app.py
app.py
(ai) Ubuntu@0151-dsm-prmx30177:~/mycode$ python app.py
Sending request to ERNIE-4.5-21B-A3B-Thinking...
```

```
Ubuntu@0151-dsm-prmx30177: ~/mycode
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Retro Synth-Wave Sunset Clock</title>
<style>
  body {
    margin: 0;
    padding: 0;
    min-height: 100vh;
    background-color: #000;
    font-family: 'Courier New', monospace;
    overflow: hidden;
  }
  .container {
    position: relative;
    height: 100vh;
  }
  .grid {
    position: absolute;
    bottom: 0;
    width: 100%;
    height: 20%;
    background:
      linear-gradient(rgba(100, 0, 200, 0.1) 1px, transparent 1px),
      linear-gradient(90deg, rgba(100, 0, 200, 0.1) 1px, transparent 1px);
    background-size: 50px 50px;
  }
  .sun {
```



Looks pretty good!



```
EXPLORER
MYCODE
  app.py
  index.html

app.py
1 import time, requests, json
2 from rich.console import Console
3 from rich.panel import Panel
4 from rich.markdown import Markdown
5
6 console = Console()
7
8 prompt = """
9 You are the newly-appointed Chief Strategy Officer of a mid-size consumer-electronics firm (US $1.1B revenue, 12 % EBITDA margin) that derives 68 % of sales from North-American big-box retailers.
10 Recent shocks:
11 (1) a 25 % tariff on Chinese-made goods taking effect in 90 days,
12 (2) a key chip-set supplier (40 % of COGS) just cancelled your contract after being acquired by a competitor,
13 (3) Best Buy informed you they will delist two of your three SKUs in 6 months unless you fund a 7 % price increase.
14
15 Board asks for a 100-day turnaround plan.
16 Deliver:
17 A. One-page executive summary (max 250 words) quantifying EBITDA at risk and target recovery.
18 B. Three strategic pillars with concrete initiatives, owners, timeline, capex needs, P&L impact (show numbers).
19 C. Clear go/no-go decision points and leading indicators we must track weekly.
20 D. Rank-order the initiatives by fastest cash-generation and by lowest execution risk; explain the trade-offs.
21 E. Stress-test the plan against a further 15 % USD appreciation and 200 bps interest-rate hike.
22 Output in tight bullet form, no marketing language, include all numbers and assumptions.
23 """
24
25 url = "http://localhost:8180/v1/chat/completions"
26
27 payload = {
28     "model": "baidu/ERNIE-4.5-21B-A3B-Thinking",
29     "messages": [{"role": "user", "content": prompt}],
30     "max_tokens": 15000
31 }
```

Next, let us do a strategy stress test for the model

```
7
8
9 3 employees, 14 % EBIT margin) that derives 68 % of sales from North-American big-box retailers.
10
11
12
```



```
Ubuntu@0151-dsm-prxmx30177: ~/mycode
(ai) Ubuntu@0151-dsm-prxmx30177:~/mycode$ python app.py
ERNIE

A. Executive Summary

EBITDA at Risk: $8.1M (current EBITDA: $12.0M) due to tariff, supplier loss, and Best Buy pressure.
Target Recovery: $10.5M (14% growth) via supply chain, pricing, and cost actions.

B. Strategic Pillars

1. Supply Chain Resilience

  • Initiative: Secure alternative chip supplier (30 days).
    • Owner: Supply Chain Director.
    • Timeline: 30 days.
    • Capex: $5M (R&D/testing for new suppliers).
    • P&L Impact:
      • High: 40% COGS recovery → $0.8M EBITDA growth.
      • Low: Partial recovery → $0.3M EBITDA loss.

2. Tariff Pass-Through & Pricing Optimization

  • Initiative: Restructure SKUs with domestic components (60 days).
    • Owner: CFO & Marketing Head.
    • Timeline: 60 days.
    • Capex: $0.
    • P&L Impact:
```

```
    • High: 7% price reduction + 10% domestic sourcing → $0.6M EBITDA stabilization.
    • Low: Partial sourcing → $0.1M EBITDA loss.

3. Best Buy Relationship & Volume Recovery

  • Initiative: Co-develop exclusive SKUs (45 days).
    • Owner: Product Management + Best Buy Liaison.
    • Timeline: 45 days.
    • Capex: $2M (marketing/inventory for new SKUs).
    • P&L Impact:
      • High: 20% volume growth → $1.0M EBITDA growth.
      • Low: Limited uptake → $0.2M EBITDA loss.
```

```
EXPLORER
... app.py x

1992 tokens in 31.90s → 62.4 tok/s
> (ai) Ubuntu@0151-dsm-prxmx30177:~/mycode$

D. Ranking & Trade-Offs

1 Fastest Cash Generation:
  • Pillar 3 (Best Buy): 45-day volume recovery → immediate cash flow.
  • Pillar 1 (Supply Chain): 30-day supplier action → long-term cost savings.
2 Lowest Execution Risk:
  • Pillar 3: No capex, high-impact negotiations.
  • Pillar 1: High capex/supplier risk but strategic criticality.
  Trade-Off: Pillar 3 prioritizes speed; Pillar 1 balances long-term risk/reward.

E. Stress-Test

  • 15% USD Appreciation:
    • Tariff cost inflation ↑ $0.7M.
    • Import costs ↑ $1.2M → Pillar 1 impact reduced to $0.5M (high) / $0.1M (low).
  • 200bps Interest-Rate Hike:
    • Financing costs ↑ $0.3M.
    • Capex ROI drops 20% → Pillar 1 capex deferred by 15 days.
  Mitigation: Accelerate Pillar 3 (no debt) and Pillar 2 (no capex).
```

We can conclude that this model is good for creative writing, coding, step-by-step thinking tasks.